Step-by-Step Prompting Template

Given a question and an agent trajectory up to the current reasoning and action steps taken by the agent to answer the question. Your task is to determine whether this most recent agent's step contains hallucinated content that could induce the question-solving process to produce an incorrect answer, and if so, explain the cause.

<Question>**{question}**</Question>
<Agent Trajectory>**{trajectory}**</Agent Trajectory>

Based on the question and agent trajectory, please predict the following:
1. Examine the agent's answer in the given <Agent Trajectory> for the provided <Question>. If the answer contains hallucinated content, please set <is_hallucinated> to True, otherwise set <is_hallucinated> to False.
2. If <is_hallucinated> is True, explain the reason that introduces or causes this hallucination. If <is_hallucinated> is False, set hallucination_reason to null.

ABSOLUTELY do not return your reasoning process. Return ONLY one JSON object that matches the format below and nothing else (in particular, never start your response with "JSON" or "Result"):

```
{
    "is_hallucination": True/False: { "type": "boolean" },
    "hallucination_reason": Reason for Hallucination { "type": ["string", "null"], "maxLength": 1000},
}
```